USN | | | | | | | | | | [23TCSE303]

THIRD SEMESTER B.Tech EXAMINATION, NOVEMBER-2024

SOFTWARE ENGINEERING

Time: 3 Hours **Maximum Marks: 100**

Instructions: Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 x2=10**
1.	**a.**	Explain prototype model working.	L2	CO1	**02**
	b.	Differentiate between scenarios and use cases.	L2	CO2	**02**
	c.	What are system design problems?	L1	CO3	**02**
	d.	What are goals of testing?	L1	CO4	**02**
	e.	Describe the four Ps of management spectrum.	L1	CO5	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Write about incremental model with block diagram.	L1	CO1	**05**
		OR			
	b.	Describe agile modeling in detail.	L2	CO1	**05**
3.	**a.**	Draw and explain the spiral view of requirement engineering process.	L3	CO2	**05**
		OR			
	b.	Write the scenario statements for the hospital management system. Take one context and at least write five statements of its working.	L1	CO2	**05**
4.	**a.**	What is system? Give an example to explain the system engineering process with a neat diagram.	L1	CO3	**05**
		OR			
	b.	Dissect the following, **i.** System integration **ii.** System installation	L4	CO3	**05**
5.	**a.**	With a neat diagram, Show the importance of inspections and testing. List the advantages of inspections.	L3	CO4	**05**
		OR			
	b.	With an example, explain the stages that the commercial software system has to go through. Describe the model of software testing process with representation.	L2	CO4	**05**
6.	**a.**	Discuss about the different stakeholders in the management spectrum.	L2	CO5	**05**
		OR			
	b.	Throw light on varieties of team coordination & communication.	L4	CO5	**05**
7.	**a.**	Explain the common-sense approach to projects.	L2	CO5	**05**
		OR			

	b.	Apply W5HH Principle to Aadhaar database management system and write the relevant points.	L3	CO5	05
		PART-C			**6 X 10=60**
8.	a.	Explain the waterfall model with its architecture? List its advantages and disadvantages.	L2	CO1	10
		OR			
	b.	Explain a unified process in detail	L2	CO1	10
9.	a.	Explain sequence model with an example. For the chosen example, illustrate sequence diagram for two context.	L2	CO2	10
		OR			
	b.	Draw the use case diagram for mental health care and patient management system. Explain requirement validation.	L3	CO2	10
10.	a.	Explain the design principles used in software engineering in detail	L4	CO3	10
		OR			
	b.	Is software evolution costly? Justify. Make your view on analysis concept, requirement analysis and analysis principles.	L5	CO3	10
11.	a.	What is regression testing? Explain its purpose, scope, automation, test suite maintenance, version control integration and bug detection.	L2	CO4	10
		OR			
	b.	With a neat diagram, explain the system procurement process including procurement issues, contractors and sub-contractors. Give an example.	L3	CO4	10
12.	a.	Who is software tester? What exactly does a software tester do? What are the skills of a Software tester?	L2	CO5	10
		OR			
	b.	What is boundary value analysis? List out the importance of it. Give an example.	L2	CO5	10
13.	a.	Considering the BMI calculator as an application given by customer to you as a requirement, Apply the SRS document structure for this application and create the document.	L3	CO6	10
		OR			
	b.	Apply test case table format to Railway track monitoring, system and write at least any five valid test cases and five invalid test cases.	L3	CO6	10